

11.1.3.2.4. Y2 Field

This field SHALL represent the position of the ending vertex along the vertical (y) axis.

11.2. Camera AV Stream Management Cluster

This cluster is used to allow clients to manage, control, and configure various audio, video, and snapshot streams on a camera. Broadly, this cluster would be used to:

1. Create and manage all streams on the camera.
2. Manage the static and dynamic characteristics of all streams on the camera.
3. Manage settings attributes that are globally applicable across fabrics and controlled by administrators.

11.2.1. Lifecycle and Resource Management

11.2.1.1. Stream Lifecycle

It is RECOMMENDED that Commissioners allocate streams when the device is commissioned, based on the expected [Stream Usages](#). Streams would therefore have very long lifetimes and only change if the expected use of the camera would change. Stream allocation is a complex and resource intensive task for devices to perform, and thus should be done as infrequently as possible.

When temporary disabling of streams is needed for things such as geo-fencing, scheduled recordings, alarms, etc; Controllers SHOULD use the [SoftRecordingPrivacyModeEnabled](#) or [SoftLivestreamPrivacyModeEnabled](#) attributes instead of deallocating and reallocating streams.

11.2.1.2. Resource Management and Stream Priorities

Allocated streams require camera resources including but not limited to CPU, memory and network bandwidth. With multiple streams allocated on the camera that can be shared among clients (potentially in different fabrics), management of these resources allocated to the streams and the stream priorities is of critical importance. The camera maintains a [ranked list of stream-usage priorities](#) that SHALL be configurable by an administrator via the [SetStreamPriorities](#) command.

11.2.1.2.1. Stream Sharing and Reuse

Whenever possible, the camera SHOULD attempt to reuse an existing stream when serving to multiple clients. If for a newly requested stream, the parameter requirements around bitrate, framerate, resolution, etc. are the same as an already existing stream, the camera SHALL reuse the existing one. For multiple available closely matching streams, one with an identical StreamUsage as the request SHOULD be selected.

If the video encoder produces a single stream, then the camera would share this stream with all receiving endpoints. However, if the video encoder is capable of producing multiple streams, the camera MAY allocate, for example, a high resolution, high bitrate stream for recording and, lower resolution, lower bitrate streams for real-time viewing on various devices.

Cameras with multiple video encoders can support simultaneous high-quality streams, with each